

Practical 3

PracticeLabAssignment

Name:

ShaileshGirishToshniwal

PRN: 202501040077

Batch: C14

3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `rows` and `columns`, representing the number of rows and the number of columns.
- The next lines contain space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

```
CODE: import numpy as np
rows, cols = map(int, input().split())

elements = []
for _ in range(rows):
    elements.extend(map(int, input().split()))

array = np.array(elements).reshape(rows, cols)

print(array)

print(array.ndim)

print(array.shape)

print(array.size)
```

Average time
0.025 s
25.25 ms

Maximum time
0.044 s
44.00 ms

2 out of 2 shown test case(s) passed
10 out of 10 hidden test case(s) passed

Test case 1
44 ms
Debug

Expected output
Actual output

3 4

3 4

1 2 3 4

1 2 3 4

5 6 7 8

5 6 7 8

9 10 11 12

9 10 11 12

[[·1··2··3··4]

[[·1··2··3··4]

·[·5··6··7··8]

·[·5··6··7··8]

·[·9·10·11·12]]

·[·9·10·11·12]]

2

2

(3,·4)

(3,·4)

12

12

Test case 2
11 ms
Debug

Expected output
Actual output

0 0

0 0

[]

[]

2

2

(0,·0)

(0,·0)

0

0

3.2.1. Numpy: Matrix Operations

The given code takes two matrices, , and , as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. Addition ()
2. Subtraction ()
3. Element-wise Multiplication ()
4. Matrix Multiplication ()
5. Transpose of Matrix A

Input Format:

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

CODE:

```
import numpy as np
```

```
#Input matrices print("Enter Matrix A:") matrix_a =  
np.array([list(map(int, input().split())) for i in range(3)])
```

```
print("Enter Matrix B:") matrix_b = np.array([list(map(int,  
input().split())) for i in range(3)])
```

```
#Addition addition_result = matrix_a +  
matrix_b print("Addition (A + B):")  
print(addition_result) # Subtraction  
subtraction_result = matrix_a - matrix_b  
print("Subtraction (A - B):")  
print(subtraction_result)
```

```
#Multiplication (element-wise) elementwise_multiplication_result  
=matrix_a * matrix_b print("Element-wise Multiplication (A *  
B):") print(elementwise_multiplication_result)
```

```
#Matrix multiplication (dot product)
```

```
matrix_multiplication_result = np.dot(matrix_a, matrix_b) print("A  
dot B:")
```

```
print(matrix_multiplication_result)  
# Transpose transpose_a = matrix_a.T  
print("Transpose of A:")  
print(transpose_a)
```

matrixOp...

Submit

Average time
0.057 s
56.75 ms

Maximum time
0.109 s
109.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 100 ms Debug

Expected output

Actual output

Enter Matrix A:

Enter Matrix A:

1 2 3

1 2 3

4 5 6

4 5 6

7 8 9

7 8 9

Enter Matrix B:

Enter Matrix B:

1 1 1

1 1 1

1 1 1

1 1 1

1 1 1

1 1 1

Addition (A + B):

Addition (A + B):

[[-2 -3 -4]

[[-2 -3 -4]

- [-5 -6 -7]

- [-5 -6 -7]

- [-8 -9 -10]

- [-8 -9 -10]

Subtraction (A - B):

Subtraction (A - B):

[[0 1 2]

[[0 1 2]

- [3 4 5]

- [3 4 5]

- [6 7 8]

- [6 7 8]

Element-wise Multiplication (A * B):

Element-wise Multiplication (A * B):

[[1 2 3]

[[1 2 3]

- [4 5 6]

- [4 5 6]

- [7 8 9]

- [7 8 9]

A dot B:

A dot B:

[[-6 -6 -6]

[[-6 -6 -6]

- [15 15 15]

- [15 15 15]

- [24 24 24]

- [24 24 24]

Transpose of A:

Transpose of A:

[[1 4 7]

[[1 4 7]

- [2 5 8]

- [2 5 8]

- [3 6 9]

- [3 6 9]

Test case 2 43 ms Debug	
Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
2 4 8	2 4 8
9 6 5	9 6 5
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
10 12 13	10 12 13
14 15 16	14 15 16
17 18 19	17 18 19
Addition (A + B):	Addition (A + B):
[[12 16 21]]	[[12 16 21]]
23 21 21	23 21 21
24 26 28]]	24 26 28]]
Subtraction (A - B):	Subtraction (A - B):
[[-8 - -8 - -5]]	[[-8 - -8 - -5]]
-5 - -9 - -11]]	-5 - -9 - -11]]
-10 - -10 - -10]]	-10 - -10 - -10]]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[-20 - -48 - 104]]	[[-20 - -48 - 104]]
126 - 90 - 80]]	126 - 90 - 80]]
119 144 171]]	119 144 171]]
A dot B:	A dot B:
[[212 228 242]]	[[212 228 242]]
259 288 308]]	259 288 308]]
335 366 390]]	335 366 390]]
Transpose of A:	Transpose of A:
[[2 9 7]]	[[2 9 7]]
4 6 8]]	4 6 8]]
8 5 9]]	8 5 9]]

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

08:24

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking: Stack the two matrices horizontally (side by side).
- Vertical Stacking: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

```
import numpy as np
```

```
# Input matrices print("Enter Array1:") arr1 =  
np.array([list(map(int, input().split())) for i in range(3)])
```

```
print("Enter Array2:") arr2 = np.array([list(map(int,  
input().split())) for i in range(3)])
```

```
#Perform horizontal stacking (hstack)  
a=np.hstack((arr1,arr2))  
print("Horizontal Stack:") print(a)
```

```
#Perform vertical stacking (vstack)  
b=np.vstack((arr1,arr2))  
print("Vertical Stack:") print(b)
```


Average time
0.052 s
52.00 ms

Maximum time
0.066 s
66.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 60 ms Debug

Expected output

Actual output

Enter Array1:
1 2 3
4 5 6
7 8 9

Enter Array1:
1 2 3
4 5 6
7 8 9

Enter Array2:
4 5 6
7 8 9
10 11 12

Enter Array2:
4 5 6
7 8 9
10 11 12

Horizontal Stack:
[[-1 -2 -3 -4 -5 -6]]
[-4 -5 -6 -7 -8 -9]
[-7 -8 -9 10 11 12]]

Horizontal Stack:
[[-1 -2 -3 -4 -5 -6]]
[-4 -5 -6 -7 -8 -9]
[-7 -8 -9 10 11 12]]

Vertical Stack:
[[-1 -2 -3]]
[-4 -5 -6]
[-7 -8 -9]
[-4 -5 -6]
[-7 -8 -9]
[-10 -11 -12]]

Vertical Stack:
[[-1 -2 -3]]
[-4 -5 -6]
[-7 -8 -9]
[-4 -5 -6]
[-7 -8 -9]
[-10 -11 -12]]

Test case 2 45 ms Debug

Expected output

Actual output

Enter Array1:
-1 -2 -3
-4 -5 -6
-7 -8 -9

Enter Array1:
-1 -2 -3
-4 -5 -6
-7 -8 -9

Enter Array2:
10 11 12
13 14 15
16 17 18

Enter Array2:
10 11 12
13 14 15
16 17 18

Horizontal Stack:
[[-1 -2 -3 10 11 12]]
[-4 -5 -6 13 14 15]
[-7 -8 -9 16 17 18]]

Horizontal Stack:
[[-1 -2 -3 10 11 12]]
[-4 -5 -6 13 14 15]
[-7 -8 -9 16 17 18]]

Vertical Stack:
[[-1 -2 -3]]
[-4 -5 -6]
[-7 -8 -9]
[10 11 12]
[13 14 15]
[16 17 18]]

Vertical Stack:
[[-1 -2 -3]]
[-4 -5 -6]
[-7 -8 -9]
[10 11 12]
[13 14 15]
[16 17 18]]

3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

```
import numpy as np
```

```
#Take user input for the start, stop, and step of the  
sequence start = int(input()) stop = int(input()) step =  
int(input())
```

```
#Generate the sequence using np.arange() n=np.arange(start,stop,  
step)
```

```
#Print the generated sequence print(n)
```

Average time
0.058 s
57.75 ms

Maximum time
0.146 s
145.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 145 ms

Debug

Expected output

Actual output

3

3

10

10

2

2

[3 5 7 9]

[3 5 7 9]

Test case 2 29 ms

Debug

Expected output

Actual output

10

10

20

20

30

30

[10]

[10]

3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

08:30

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i \text{ OR } B_i$).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

```
import numpy as np
```

```
def array_operations(A, B):
```

```
    # Convert A and B to NumPy arrays
```

```
    A = np.array(A)
```

```
    B = np.array(B) #
```

```
    Arithmetic
```

```
    Operations
```

```
    sum_result = A +
```

```
    B diff_result = A
```

```
    -B prod_result =
```

```
    A * B
```

```
    # Statistical Operations
```

```
    mean_A = np.mean(A)
```

```
    median_A = np.median(A)
```

```
    std_dev_A = np.std(A)
```

```
    # Bitwise Operations
```

```
    and_result = A & B    or_result =
```

```
    A|B    xor_result = A ^ B
```

```
    #Output results with one space between each element    print("Element-
wiseSum:", ''.join(map(str, sum_result)))    print("Element-wise
```

```
Difference:", ''.join(map(str, diff_result)))    print("Element-wise Product:", '
'.join(map(str, prod_result)))
```

```
    print(f"Mean of A: {mean_A}")
print(f"Median of A: {median_A}") print(f"Standard
Deviation of A: {std_dev_A}")
```

```
    print("Bitwise AND:", ''.join(map(str, and_result)))
print("Bitwise OR:", ''.join(map(str, or_result)))    print("Bitwise
XOR:", ''.join(map(str, xor_result)))
```

```
A= list(map(int, input().split())) # Elements of array A
B= list(map(int, input().split())) # Elements of array B
array_operations(A, B)
```

The screenshot displays a code execution interface with the following components:

- Performance Metrics:**
 - Average time: 0.065 s (65.33 ms)
 - Maximum time: 0.143 s (143.00 ms)
- Test Results:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case Details (Test case 1, 143 ms):**

Expected output	Actual output
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32
Mean of A: 2.5	Mean of A: 2.5
Median of A: 2.5	Median of A: 2.5
Standard Deviation of A: 1.118033988749895	Standard Deviation of A: 1.118033988749895
Bitwise AND: 1 2 3 0	Bitwise AND: 1 2 3 0
Bitwise OR: 5 6 7 12	Bitwise OR: 5 6 7 12
Bitwise XOR: 4 4 4 12	Bitwise XOR: 4 4 4 12

3.2.5. Numpy: Copying and Viewing Arrays

04:23

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`. □ Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>

Copy array: <copy_array>

```
import numpy as np
```

```
inputlist = list(map(int,input().split(" ")))
```

```
#Original array original_array =
```

```
np.array(inputlist)
```

#Create a view

```
view_array=original_array.view()
```

#Create a copy copy_array

```
=original_array.copy()
```

```
#Modify the view view_array[0] = 99 print("Original array  
after modifying view:", original_array) print("View array:",  
view_array)
```

```
#Modify the copy copy_array[1] = 88 print("Original array  
after modifying copy:", original_array) print("Copy array:",  
copy_array)
```

Average time 0.018 s 18.50 ms Maximum time 0.032 s 32.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 32 ms Debug	
Expected output	Actual output
10 20 30 40 50 60 70 80	10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]	Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]	View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]	Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]	Copy array: [10 88 30 40 50 60 70 80]

Test case 2 17 ms Debug	
Expected output	Actual output
99 2 3 4 5	99 2 3 4 5
Original array after modifying view: [99 2 3 4 5]	Original array after modifying view: [99 2 3 4 5]
View array: [99 2 3 4 5]	View array: [99 2 3 4 5]
Original array after modifying copy: [99 2 3 4 5]	Original array after modifying copy: [99 2 3 4 5]
Copy array: [99 88 3 4 5]	Copy array: [99 88 3 4 5]

3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

05:03

The given code in the editor takes a single array, `array1`, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- `search_value`: The value to search for in the array.
- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. Searching: Find the indices where `search_value` appears in `array1` and print these indices.
2. Counting: Count how many times `count_value` appears in `array1` and print the count.
3. Broadcasting: Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
4. Sorting: Sort `array1` in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.
3. An integer `count_value` represents the value to count in the array.
4. An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

1. The indices where `search_value` occurs in `array1`.
2. The count of occurrences of `count_value` in `array1`.
3. The array after adding the `broadcast_value` to each element.
4. The sorted array.

```
import numpy as np
```



```
#Input array from the user array1 =  
np.array(list(map(int, input().split())))
```

```
#Searching search_value = int(input("Value  
to search: ")) count_value = int(input("Value  
to count: ")) broadcast_value =  
int(input("Value to add: "))
```

```
#Find indices where value matches in array1  
a=np.where(array1==search_value)[0] print(a)  
# Count occurrences in array1 b=np.count_nonzero(array1==count_value)  
print(b) # Broadcasting  
addition c=array1 +  
broadcast_value print(c) #  
Sort the first array  
d=np.sort(array1) print(d)
```

Average time

0.035 s

34.75 ms

Maximum time

0.061 s

61.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

61 ms

Debug

Expected output

1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0 1 2]

3

[3 3 3 4 4 4]

[1 1 1 2 2 2]

Actual output

1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0 1 2]

3

[3 3 3 4 4 4]

[1 1 1 2 2 2]

Test case 2

27 ms

Debug

Expected output

1 2 3 4 5

Value to search: 6

Value to count: 6

Value to add: 6

[]

0

[-7 -8 -9 -10 -11]

[1 2 3 4 5]

Actual output

1 2 3 4 5

Value to search: 6

Value to count: 6

Value to add: 6

[]

0

[-7 -8 -9 -10 -11]

[1 2 3 4 5]

3.2.7. Student Data Analysis and Operations

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.
- Find total students: Determine the total number of students in the dataset.
- Print all student roll numbers: Extract and print the roll numbers of all students.
- Print Subject 1 marks: Extract and print the marks of all students in Subject 1.
- Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.
- Find maximum marks in Subject 3: Identify the highest marks in Subject 3.
- Print all subject marks: Display the marks of all students for each subject.
- Find total marks of students: Compute the total marks for each student across all subjects.
- Find the average marks of each student: Compute the average marks for each student.
- Find average marks of each subject: Compute the average marks for all students in each subject.
- Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.
- Find average marks of Subject 1 and Subject 3: Compute the average marks for Subject 1 and Subject 3.
- Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.
- Find the roll number of the student with minimum marks in Subject 2: Identify the student with the lowest marks in Subject 2 and print their roll number.
- Find the roll number of students who scored 24 marks in Subject 2: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- Find the count of students who got less than 40 marks in Subject 1: Count the number of students who scored less than 40 marks in Subject 1.
- Find the count of students who got more than 90 marks in Subject 2: Count the number of students who scored more than 90 marks in Subject 2.
- Find the count of students who scored ≥ 90 in each subject: Count the number of students who scored 90 or more marks in each subject.
-

- Find the count of subjects in which each student scored ≥ 90 : Determine how many subjects each student scored 90 or more marks in.
- Print Subject 1 marks in ascending order: Sort and print the marks of students in Subject 1 in ascending order.
- Print students who scored between 50 and 90 in Subject 1: Display students who scored marks between 50 and 90 in Subject 1.
- Find index positions of students who scored 79 in Subject 1: Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

CODE:

```
import numpy as np
```

```
a= np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
```

```
#1. Print all student details print("All
student Details:\n", a )
```

```
#2. print total students
```

```
print("Total Students:",len(a))
```

```
#3. Print all student Roll numbers print("All
Student Roll Nos", a[:,0] )
```

```
#4. Print subject 1 marks
print("Subject 1 Marks", a[:,1])
```

#5. print minimum marks of Subject 2 print("Min
marks in Subject 2", np.min(a[:, 2]))

#6. print maximum marks of Subject 3 print("Max
marks in Subject 3", np.max(a[:,3]))

#7. Print All subject marks print("All
subject marks:", a[:,1:])

#8. print Total marks of students print("Total
Marks", np.sum(a[:,1:], axis = 1))

#9. print average marks of each student print(np.round(np.mean(a[:,1:],
axis = 1),1))

#10. print average marks of each subject print("Average Marks of
each subject", np.mean(a[:, 1:], axis = 0))

#11. print average marks of S1 and S2 print("Average Marks of S1
and S2", np.mean(a[:, 1:3],axis=0))

#12. print average marks of S1 and S3 print("Average Marks of S1 and S3",
np.mean(a[:, [1, 3]], axis = 0))

#13. print Roll number who got maximum marks in Subject 3 max_sub3_index =
np.argmax(a[:, 3]) print("Roll no who got maximum marks in Subject 3",
a[max_sub3_index , 0]) # 14. print Roll number who got minimum marks in

```
Subject 2 min_sub2_index = np.argmin(a[:, 2]) print("Roll no who got minimum  
marks in Subject 2", a[min_sub2_index, 0] )
```

#15. print Roll number who got 24 marks in Subject 2

```
print(f"Roll no who got 24 marks in Subject 2 [{a[a[:,2] == 24][:,0]}]" )
```

#16. print count of students who got marks in Subject 1 < 40 count_sub1_lt_40 =
np.sum(a[:,1]<40) print("Count of students who got marks in Subject 1 < 40",
count_sub1_lt_40)

#17. print count of students who got marks in Subject 2 > 90 count__sub2_gt_90 =
np.sum(a[:,2]>90) print("Count of students who got marks in Subject 2 > 90:",
count__sub2_gt_90)

#18. print count of students in each subject who got marks >= 90 count_each_subject_90plus
=np.sum(a[:,1:]>=90 , axis = 0)
print("Count of students in each subject who got marks >= 90:", count_each_subject_90plus
)

#19. print count of subjects in which each student got marks >= 90 print("Roll
no:", a[:,0].astype(float))
print("Count of subjects in which student got marks >= 90:", np.sum(a[:,1:] >= 90 , axis = 1)
)

#20. Print S1 marks in ascending order print(np.sort(a[:,1]))

21. Print S1 marks ≥ 50 and ≤ 90

```
s1_filteredv = a[(a[:,1]>=50) & (a[:,1]<=90)]
```

```
print(s1_filteredv) print(a)
```

#22. Print the index position of marks 79

```
indices_79 = np.where(a[:,1:]==79)[0] print((indices_79,))
```

Average time
0.103 s
103.00 ms

Maximum time
0.103 s
103.00 ms

1 out of 1 shown test case(s) passed

Test case 1 103 ms Debug

Expected output

Actual output

All student Details:
[[301, 67, 77, 88.]]
[[302, 78, 88, 77.]]
[[303, 45, 56, 89.]]
[[304, 88, 98, 45.]]
[[305, 78, 88, 99.]]
[[306, 68, 78, 36.]]
[[307, 79, 12, 45.]]
[[308, 46, 45, 77.]]
[[309, 89, 32, 88.]]
[[310, 79, 24, 33.]]
Total Students: 10
All Student Roll Nos [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]
Subject 1 Marks [67, 78, 45, 88, 78, 68, 79, 46, 89, 79.]
Min marks in Subject 2 12.0
Max marks in Subject 3 99.0
All subject marks: [[67, 77, 88.]]
[[78, 88, 77.]]
[[45, 56, 89.]]
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